

# The Postulates of Quantum Mechanics

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# The Postulates of Quantum Mechanics

1. Each system can be characterized by a wave function, which may be complex,

$$\Psi(\vec{r}, t)$$

and which contains all the information that is known about the system.

# Postulates

2. The wave function is defined such that

$$p(\vec{r}, t)dV = \Psi^*(\vec{r}, t)\Psi(\vec{r}, t)dV = |\Psi(\vec{r}, t)|^2 dV$$

is the probability of finding the system in the volume element  $dV$ . The consequence of this is that

$$\int \Psi^*(\vec{r}, t)\Psi(\vec{r}, t)dV = 1$$

i.e. The wave function is normalized

# Postulates

3. With every dynamical variable, there are associated operators

|                 |                                                        |
|-----------------|--------------------------------------------------------|
| $\vec{r}$       | $\vec{r}_{op} = \vec{r}$                               |
| $\vec{p}$       | $\vec{p}_{op} = -i\hbar\vec{\nabla}$                   |
| $\varepsilon$   | $\varepsilon_{op} = i\hbar\frac{\partial}{\partial t}$ |
| $B(\vec{r}, t)$ | $B_{op} = B(\vec{r}, -i\hbar\vec{\nabla})$             |

# Postulates

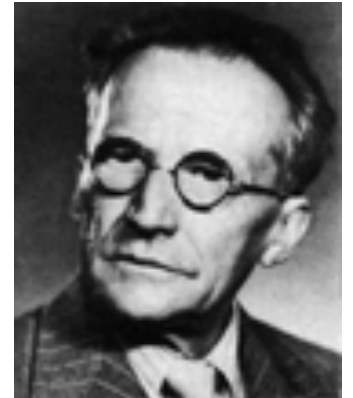
4. The expectation value of any physical observable is

$$\langle B \rangle = \int \Psi^*(\vec{r}, t) B_{op} \Psi(\vec{r}, t) dV$$

The wave function is determined by using Schrodinger's equation.

# Finding the Wave Function

- We will obtain the wave function by solving Schrodinger's wave equation.
- Schrodinger obtained the equation by starting with the equation for the total energy of a particle and applying the operators for each term.



# Energy of a Particle

- The total energy of a particle is the sum of its kinetic and potential energies.
- We can write the kinetic energy as

$$\text{K.E.} = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

- The form of the potential energy will depend on the nature of the particle. For now we will call it  $V(x,t)$  as it may be a function of both space and time.

# Hamiltonian

- In classical mechanics, the sum of the potential and kinetic energy is called the Hamiltonian.

$$H = \frac{p^2}{2m} + V(\vec{r}, t) = \varepsilon$$

- Substituting the operators for  $p$  and  $\varepsilon$ ,

$$i\hbar \frac{\partial \Psi(t, \vec{r})}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \Psi(t, \vec{r}) + V(t, \vec{r}) \Psi(t, \vec{r})$$



# Why Schrodinger chose this form

- He needed a linear equation to describe the propagation of wave packets,
- He wanted the equation to be symmetric with the classical conservation of energy statement.

# What does the Wave Equation Describe?

- It describes the time dependent, transient behavior (motion) of any collection of quantum particles subject to body forces, including internal motion, collisions and chemical reactions.

# Analytic Solutions of the Wave Equation we will Explore:

- Separate time dependence from stationary behavior
- Two Particle Systems
- Particle in a box
- Rigid Rotator
- Harmonic Oscillator
- Hydrogenic Atom

We will also explore using numerical methods for real systems.